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MASTER COMPU PHYS - FORTRAN POOL

Numerical project:
Heat diffusion in homogeneous and inhomogeneous media

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M1 CompuPhys

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1 Introduction

Heat diffusion plays a crucial role in determining temperature distribution and thermal behavior in metallic bars. In this report, the numerical simulation of heat diffusion in a homogeneous copper bar is presented. The aim of the simulation was to study the temperature distribution along the bars over time and to analyze the effect of different boundary conditions.

The simulation was carried out using the finite-difference method and a program written in Fortran. The heat diffusion equation was solved using numerical integration techniques, namely, the Forward Euler and Backward Euler schemes. The simulation was performed with Dirichlet, Neumann, and Robin boundary conditions. The results of the simulation were compared with experimental measurements on a copper bar, and the accuracy of the simulation was analyzed.

The report is organized as follows: First, the theoretical background of heat diffusion and the heat diffusion equation are introduced. Then, the methods and techniques used in the simulation are described in detail. The results of the simulation are presented and analyzed, and the comparison with experimental data is discussed. Finally, the report concludes with a summary of the findings.

The purpose of this report is to provide a comprehensive understanding of the numerical simulation of heat diffusion in metallic bars, and to demonstrate the usefulness of numerical simulations in the study of heat transfer processes.

2 Theory

In this section, we will provide a brief overview of the mathematical and physical principles behind the program.

The heat flux density is denoted by $\vec{\Phi}(\vec{r}, t)$ in Wm^{-2} . Fourier's law states that

$$\vec{\Phi}(\vec{r}, t) = -\lambda \vec{\nabla} T(\vec{r}, T) \quad (1)$$

where T is the temperature and λ (sometimes denoted k) is the thermal conductivity.

The Heat diffusion equation For diffusion along a single direction for a homogeneous medium reads as :

$$\frac{\partial}{\partial t} T(x, t) = \alpha \frac{\partial^2}{\partial x^2} T(x, t) + \tilde{f}(x, t) \quad (2)$$

where $\alpha = \lambda/(\rho c_p)$ is the thermal diffusivity, c_p is the specific heat capacity and ρ is the density.

The source and sink term is defined as $\tilde{f}(x, t) = f(x, t)/(\rho c_p)$ where $f(x, t)$ is the amount of heat lost or received per unit time and per unit volume (the SI unit of f is Wm^{-3}).

We are interested in solving this equation numerically thus it's useful to reduce the units as follows:

Let T_0 be a characteristic temperature, L_0 a characteristic length and t_0 a characteristic time. In terms of the reduced quantities we get:

$$T^* = T/T_0 \quad x^* = x/L_0 \quad t^* = t/t_0$$

The heat equation becomes

$$\frac{\partial}{\partial t^*} T^* = \alpha^* \frac{\partial^2}{\partial x^{*2}} T^* + \tilde{f}^* \quad (3)$$

where,

$$\alpha^* = \alpha \frac{t_0}{L_0^2}$$

and,

$$\tilde{f}^* = \frac{t_0}{\rho c_p T_0} f = \frac{t_0 \alpha}{T_0 \lambda} f$$

are dimensionless coefficients.

one can always set $\alpha^* = 1$, by choosing suitably the unit of length L_0 and the unit of time t_0 . The program will use reduced units in calculations.

Integration Scheme

To solve the heat equation numerically, the domain is discretized into a grid with $N + 1$ points and grid spacing Δx . The time is also discretized into a grid with $M + 1$ points and grid spacing Δt .

The first derivative of a function $u(x)$ can be estimated numerically by one of the following formulae

$$\left. \frac{\partial u}{\partial x} \right|_i \approx \begin{cases} \frac{u_{i+1} - u_i}{\Delta x} & \text{(forward finite difference)} \\ \frac{u_i - u_{i-1}}{\Delta x} & \text{(backward finite difference)} \\ \frac{u_{i+1} - u_{i-1}}{2\Delta x} & \text{(centred finite difference of step } 2\Delta x) \\ \frac{u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}}}{\Delta x} & \text{(centred finite difference of step } \Delta x \text{ involving off-grid points)} \end{cases} \quad (4)$$

The **Forward Euler** scheme is used to approximate the solution at the next time step based on the current solution. The scheme is derived from equation 4 and is given by:

$$u_i^{n+1} = u_i^n + r(u_{i-1}^n - 2u_i^n + u_{i+1}^n) + \tilde{f}_i^n \Delta t \quad i = 1, \dots, N - 1 \quad (5)$$

where u_i^n is the temperature at point i at time step n , $\tilde{f} = f/(\rho c_p) = f\alpha/\lambda$, $r = \alpha\Delta t/(\Delta x)^2$, $n = 0, \dots, M$ and $i = 1, \dots, N - 1$. For this scheme to converge, r must be less than 0.5.

The **Backward Euler** scheme is similar, but the solution at the next time step is based on the updated solution. The scheme is given by:

$$u_i^{n+1} - r(u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1}) = u_i^n + \tilde{f}_i^{n+1} \Delta t \quad i = 1, \dots, N - 1 \quad (6)$$

It should be noted that the Backward Euler scheme is stable for any value of r , while the stability of the Forward Euler scheme is dependent on the value of r , it is only stable for $r \leq 0.5$.

Boundary conditions must also be considered in the numerical solution. Three types of boundary conditions will be used in this report: Dirichlet, Neumann, and Robin boundary conditions.

Dirichlet boundary conditions are defined by fixed values for the function at the end points. For example, at $x = 0$, u_0 is fixed at a certain value. Similarly, at $x = L$, u_N is fixed at a certain value.

Neumann boundary conditions are defined by fixed values for the gradient of the function at the end points. For example, at $x = 0$, $\left. \frac{\partial T}{\partial x} \right|_{x=0}$ is fixed at a certain value. Similarly, at $x = L$, $\left. \frac{\partial T}{\partial x} \right|_{x=L}$ is fixed at a certain value.

Robin boundary condition relates the value of a function and its normal derivative at a boundary through a linear combination of the two. For example, at $x = L$, the heat flux is given by $\phi(L, t) = -\lambda \left. \frac{\partial T}{\partial x} \right|_{x=L} = h(T(L, t) - T_{room})$ for all t , where h (in $\text{W m}^{-2} \text{K}^{-1}$) is the convection coefficient at the bar-air interface.

3 System

A bar made of Cu is used as the medium for heat diffusion. The bar can be modeled as a grid with $N+1$ points and grid spacing Δx in 1D. The temperature of the bar is discretized on a grid with $M+1$ points and grid spacing Δt . The bar is subjected to heat inputs and external conditions at its boundaries, which drive the diffusion of heat through the bar over time. The aim of the experiment is to study the heat diffusion in the bar, as well as the effect of the material properties Cu on the diffusion process.

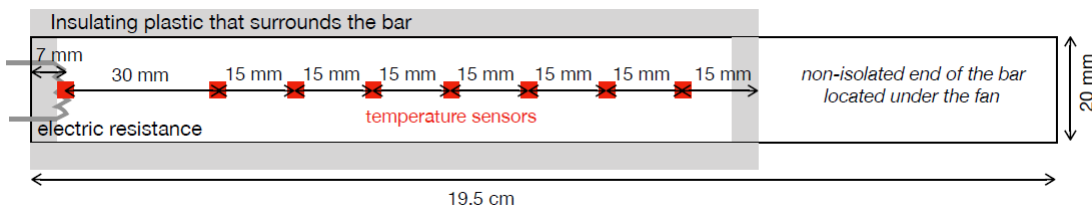


Figure 1: Top view of the bar showing its dimensions, its isolation, and the positions of the temperature sensors. The bar volume is $195 \times 20 \times 10 \text{ mm}^3$.

4 Code

```

1  ...
2  !-----Simulation-----!
3  method = 1 !can be switched between 0 to use the Forward Scheme or 1 for the Backward scheme
4
5  do j= 1, N+1
6      u(1,j) = 1
7  enddo
8
9  do t = 1, M
10
11      if (method==1) then
12
13          !----Backward Euler----!
14          BC_x0 = 1 ! 0 for Dirichlet, 1 for Neuman
15          BC_xL = 2 ! 0 for Dirichlet, 1 for Neuman, 2 for Robin
16
17          !parameter for the boundary conditions
18          du_0 = 0
19          du_1 = -0.098905
20
21          ! Simulate the Temperature
22
23          call Backward_Euler(r,N,dx,du_0,du_1,du_N,u,t,cdu,BC_x0, BC_xL)
24
25          !-----!
26
27          else if (method == 0) then
28              BC_x0 = 1 ! 0 for Dirichlet, 1 for Neuman
29              BC_xL = 0 ! 0 for Dirichlet, 1 for Neuman
30
31              !parameter for the boundary conditions
32              du_0 = -0.098905
33              du_N = 0
34
35              ! Source/Sink Term
36              do j = 1, N+1
37                  f_EM_radiation(t,j) = (time0*alphaCu/(T0*lambdaCu)) * area*sigma*(u(t,j)**4 - T0**4)/&
38                      (bar_height*bar_length*bar_width)
39              enddo
40
41              ! choose between No sink Term or With EM radiation
42              ! the default is without
43              f = 0
44              !f = f_EM_radiation
45
46              !-----Forwad Euler-----!
47              if (BC_x0 == 0) then ! Dirichlet at x=0
48                  call Dirichlet_0(u, t)
49
50              else if(BC_x0 == 1) then ! Neuman at x=0
51                  call Neuman_0(u, du_0, r, dx, dt, f, t)
52              endif
53
54              if (BC_xL == 0) then ! Dirichlet at x=L
55                  call Dirichlet_L(u, t, N)
56
57              else if(BC_xL == 1) then ! Neuman at x=L
58                  call Neuman_L(u, N, du_N, r, dx, dt, f, t)
59              endif
60
61              ! Euler: x= deltaX ---- N-1
62              do j = 2, N
63                  call euler(u, r, dt, f, t, j)
64              end do
65
66          end if
67
68      end do
69      ...

```

Listing 1: Code Snippet for 1D Heat Transfer Simulation (heat_1D.f90)

The main code used in this simulation is the heat_1d.f90 script, which allows for the simulation of 1D heat transfer in a bar made of copper material. It utilizes the Forward and Backward Euler schemes to solve the heat equation. The

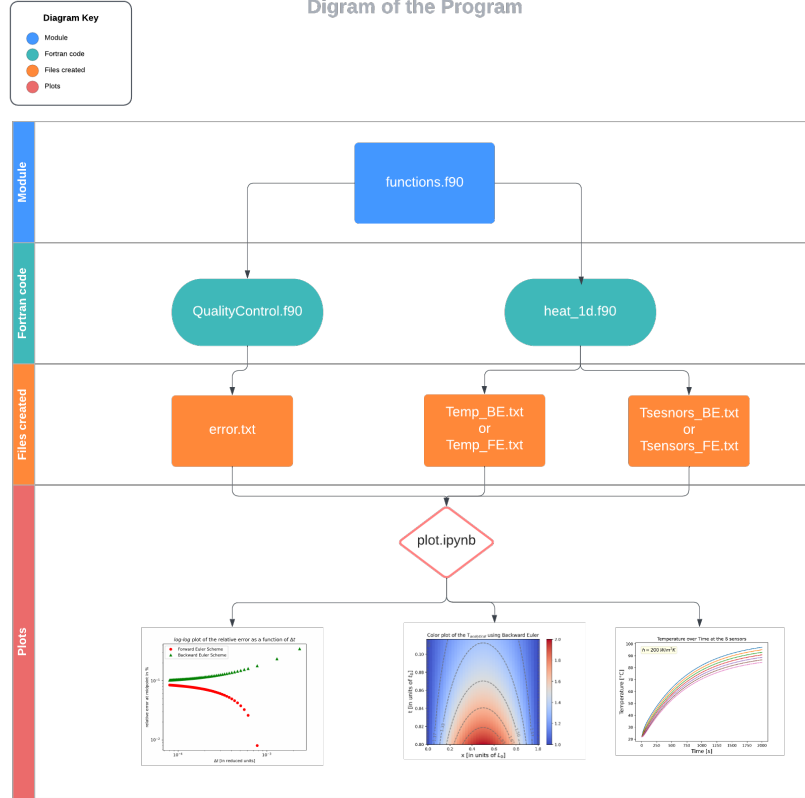


Figure 2: Program Structure Diagram

dimensions of the bar are defined as `bar_length`, `bar_width`, and `bar_height`, while the thermal diffusivity and thermal conductivity of the copper material are represented by `alphaCu` and `lambdaCu`, respectively. The structure of the various scripts used in the simulation is presented in Figure 2.

The simulation parameters, `M` and `N`, indicate the number of discretized points in time and space, respectively. The left boundary temperature is given as a Neumann boundary condition with a constant heat flux of 60000 W/m^2 . The temperature at the right boundary is defined using either a Dirichlet, Neumann or Robin boundary condition. The simulation runs for a total of 2000 seconds.

The code begins by printing information about the system and the simulation parameters, followed by allocating memory for the temperature array "`u`" and initializing its values using a uniform temperature of 1 (corresponding to 295 K in reduced units). The Forward Euler or Backward Euler method is then applied to simulate the temperature profile over time. The results are saved to a text file and can be plotted using the provided python script ("`plot.ipynb`").

The behavior of the code can be controlled by changing the values of the parameters `method`, `Bc_x0`, and `Bc_xL`. These parameters determine the solution method used, the initial conditions, and the boundary conditions, respectively.

`method`: This parameter can be set to "0" to use the Forward Euler method or "1" to use the Backward Euler method.
`Bc_x0`: This parameter defines the boundary condition at $x = x_0$, where x_0 represents the initial position of the bar. It can be set to "0" for a Dirichlet boundary condition or "1" for a Neumann boundary condition.
`Bc_xL`: This parameter specifies the boundary condition at $x = x_L$, where x_L represents the final position of the bar. It can be set to "0" for a Dirichlet boundary condition, "1" for a Neumann boundary condition, or "2" for a Robin boundary condition (when the Backward Euler method is used).

By adjusting the values of these parameters, the code can be run in various configurations to generate different solutions to the heat transfer problem.

5 Quality Control: Testing with Analytical Solutions

One of the critical aspects of conducting experiments and simulations is ensuring the accuracy of the results obtained. In this section, we aim to verify the validity of our approach by comparing the results from our simulations with known analytical solutions. Analytical solutions provide an established benchmark to which the results of our simulations can be compared.

We used the following analytical solution of the diffusion equation:

$$T_{analytic}(x, t) = 1 + \sin(\pi x) \exp(-\alpha \pi^2 t) \quad (\text{in reduced units}) \quad (7)$$

with an initial condition of $T_{initial}(x) = 1 + \sin(\pi x)$ and Dirichlet boundary conditions of $T(0, t) = T(L, t) = 1$ (for all t). Here, α is the reduced thermal diffusivity, chosen to be 1 ($= \alpha t_0 / L_0$).

To evaluate the accuracy of our numerical solution, we compute the relative error:

$$\epsilon = |T - T_{analytic}| / T_{analytic} \quad (8)$$

at the mid-point, $x = 1/2$, when the temperature of this point should have reached the value $3/2$ according to the analytic solution, i.e. after a simulation time τ defined by the condition $T(1/2, \tau) = 3/2$.

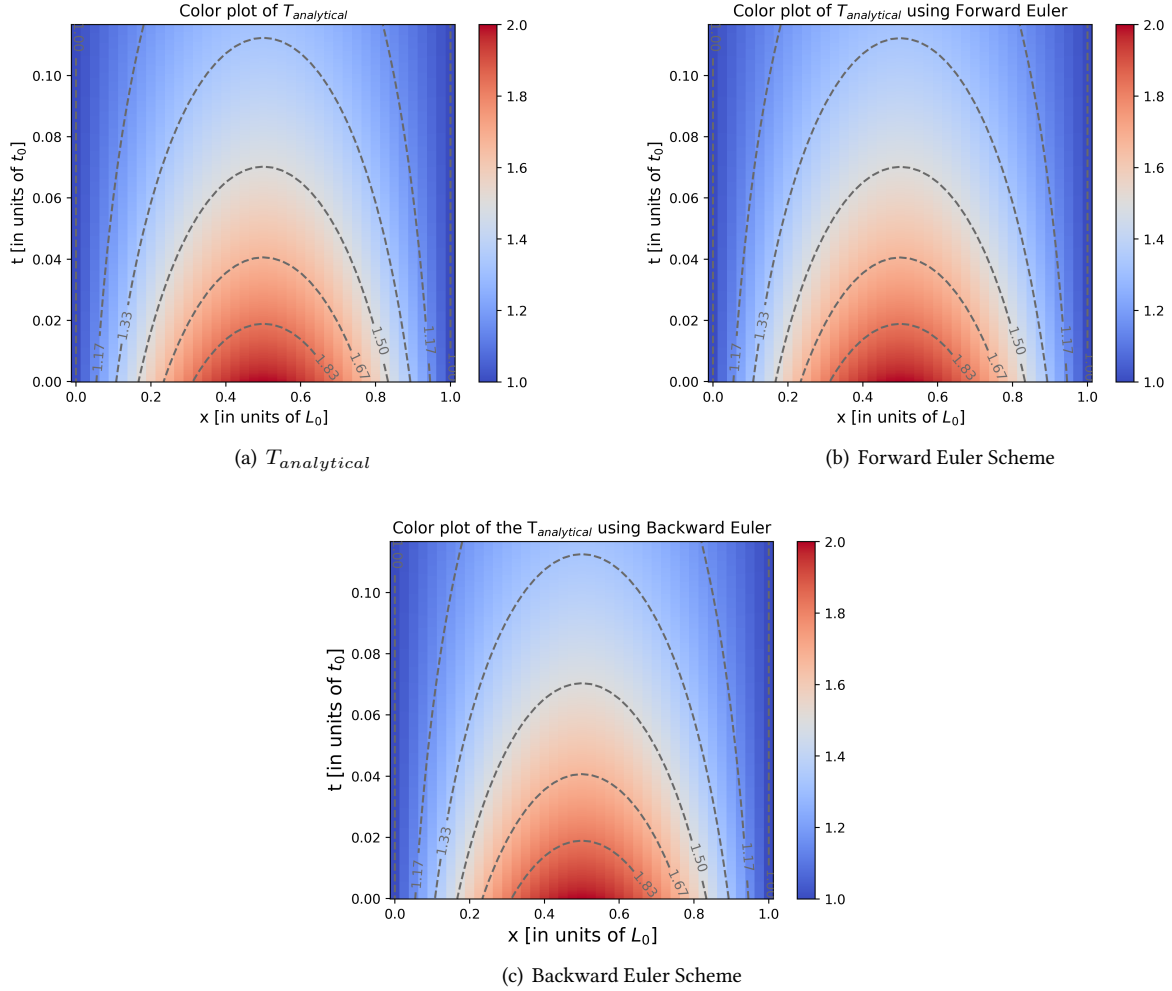


Figure 3: Temperature profile (in reduced units, when $\alpha = 1$) along the Cu bar as function of time according to (a) the analytical solution (7), (b) the Forward Euler and (c) the Backward Euler schemes. The temperatures are shown by a color scale ranging from $T = 1$ (blue) to $T = 2$ (red).

In Figure 3, we present a color plot of the temperature profile according to the analytical solution and the Forward and Backward Euler schemes. The plot was created using $L_0 = \text{length of the bar} = 0.195m$, $t_0 = L_0^2 / \alpha_{Cu}$, $\Delta t = 2.92 \cdot 10^{-4}s$ ($M = 20000$) and $\Delta x = 2.5 \cdot 10^{-2}m$ ($N = 40$). The temperature is shown by a color scale ranging from $T = 1$ (blue) to $T = 2$ (red) in reduced units ($\alpha = 1$).

The plot clearly shows that the Forward and Backward Euler schemes produce temperature profiles that are very similar to the analytical solution.

In Figure 4, we present a log-log plot comparing the error at the midpoint ($x = 0.5$) using the Forward and Backward Schemes. The relative error is on the y-axis, and the change in Δt is on the x-axis, both in logarithmic scales. The Δx was kept fixed at $1/N$ with $N = 14$, $\Delta x = 1/14 = 0.0714$ in reduced units.

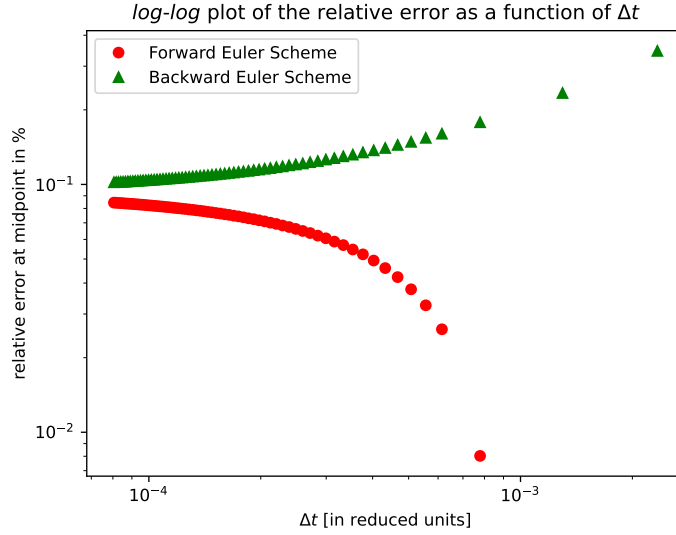


Figure 4: *log-log* plot comparing the error at the midpoint ($x = 0.5$) using the Forward and Backward Scheme

The red line represents the forward Euler solution, and the green line represents the backward Euler solution. The plot shows that as Δt increases, the relative error produced by the forward Euler solution decreases, indicating that the forward Euler solution becomes more accurate with larger time steps. On the other hand, the relative error produced by the backward Euler solution increases, indicating that the backward Euler solution becomes less accurate with larger time steps.

6 Simulation Results

In this section, we present the results of our numerical simulation of the temperature distribution in the bar using both Forward Euler and Backward Euler schemes. We compare the simulated temperature profiles with the experimental data obtained from the 8 sensors positioned along the bar. The simulation parameters used for both methods are:

$$L_0 = 0.195 \text{ m}, \quad t_0 = \frac{L_0^2}{\alpha_{Cu}}, \quad T_0 = 295 \text{ K}, \quad \Delta t = 2.92 \times 10^{-4} \text{ s} (M = 20000), \quad \Delta x = 2.5 \times 10^{-2} \text{ m} (N = 40)$$

$$\text{initial heat flux: } \phi_0 = \text{POWER} \div \text{AREA} = 12 \text{ W} \div 0.02 \times 0.01 \text{ m}^2 = 60000 \text{ Wm}^{-2}$$

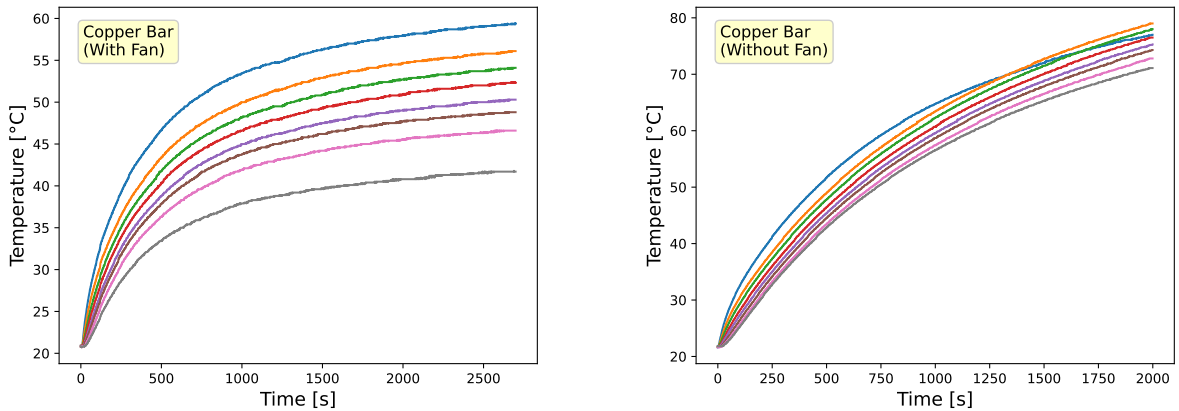


Figure 5: Experimental results for the temperature of at the 8 sensors for a copper bar. The heated on one side by a power of 12 W; In the first plot, a fan cools down the other end of the bar, whereas this fan is switched off for the second plot.

6.1 Forward Euler Method

We first consider the case of the Forward Euler scheme, with two cases: the fan turned off, and the fan turned on. For each case, we present a color plot of the temperature profile along the bar, and a plot of the temperature at the 8 sensors to compare it with the experimental data.

Case 1: Fan turned off

In this case, we will use Neumann boundary conditions at $x = 0$ and $x = L$ to account for the heat flux. The heat flux at $x = 0$ is given by $\phi_0 = -\lambda \frac{\partial T}{\partial x} = 60000 \text{ W m}^{-2}$, where $\lambda = 401 \text{ W m}^{-1} \text{ K}^{-1}$ is the thermal conductivity of copper. The heat flux at $x = L$ is set to $\phi_L = 0$.

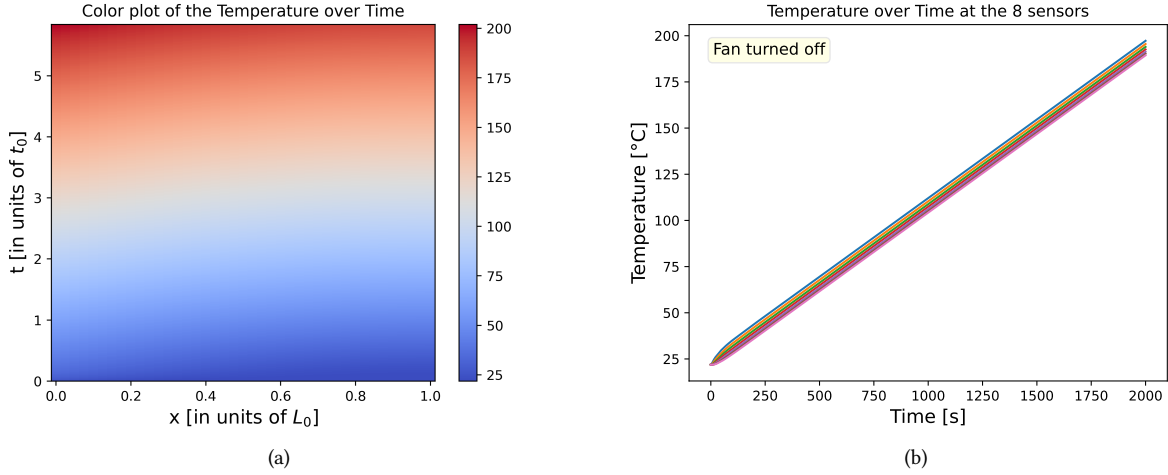


Figure 6: (a) Temperature profile along the Cu bar as a function of time (fan turned off) according to the simulation results using the forward Euler scheme. The temperatures are shown by a color scale ranging from $T = 22^\circ \text{C}$ (blue) to $T = 200^\circ \text{C}$ (red). (b) The temperature as a function of time at the 8 sensors.

The color plot of the temperature profile along the Cu bar as a function of time, shown in Figure 6(a), indicates that when the fan is turned off, the temperature profile gradually increases from one end to the other due to conduction, with slightly higher temperatures closer to the start of the bar.

Figure 6(b) shows a plot of the temperature at the 8 sensors as a function of time, comparing the simulation results with the experimental data (Figure 5). The plot shows that there is a big disparity between the simulation and the experimental data, the temperature is linearly increasing with time, whereas the experimental data show a slight curvature, and the temperature also reaches a very high value of 200°C compared to 80°C in the experimental data. This suggests that this approach does not produce accurate results for the case of the fan turned off, likely due to neglecting phenomena such as heat convection and black body radiation.

To account for the effect of black body radiation in the code, we can set $f = f_{\text{EM_radiation}}$ instead of 0 in the code (1). The reduced sink term is then calculated as follows:

$$\tilde{f}^* = \frac{t_0}{\rho c_p T_0} f = \frac{t_0 \alpha}{T_0 \lambda} f = \frac{t_0 \alpha}{T_0 \lambda} \frac{\dot{Q}_{\text{radiation}}}{V} = \frac{t_0 \alpha}{T_0 \lambda} \frac{A \sigma (T^4 - T_{\text{ambient}}^4)}{V} \quad (9)$$

where, $A = 2(L \times h) + 2(L \times w) + 2(w \times h)$, $V = L \times h \times w$, $T_{\text{ambient}} = T_0 = 295 \text{ K}$, and $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Figure 7(a) shows the temperature profile along the Cu bar as a function of time when black body radiation is included in the simulation, while Figure 7(b) shows the temperature at the 8 sensors. No changes on the shape of the temperature lines occurs, however, the maximum temperature reached decrease from 200°C to 120°C . Although this is still far from the experimental data maximum of 80°C , These figures show improved agreement between the simulation results and the experimental data, indicating the importance of accounting for black body radiation in the simulation.

Case 2: Fan turned on

In this case, we will use Neumann boundary condition at $x = 0$ to account for the heat flux, and Dirichlet boundary condition at $x = L$ to keep the temperature constant. The heat flux at $x = 0$ is the same as before given by $\phi_0 = -\lambda \frac{\partial T}{\partial x} = 60000 \text{ W m}^{-2}$. The temperature at $x = L$ is set to $T_L = T_0 = 295 \text{ K}$.

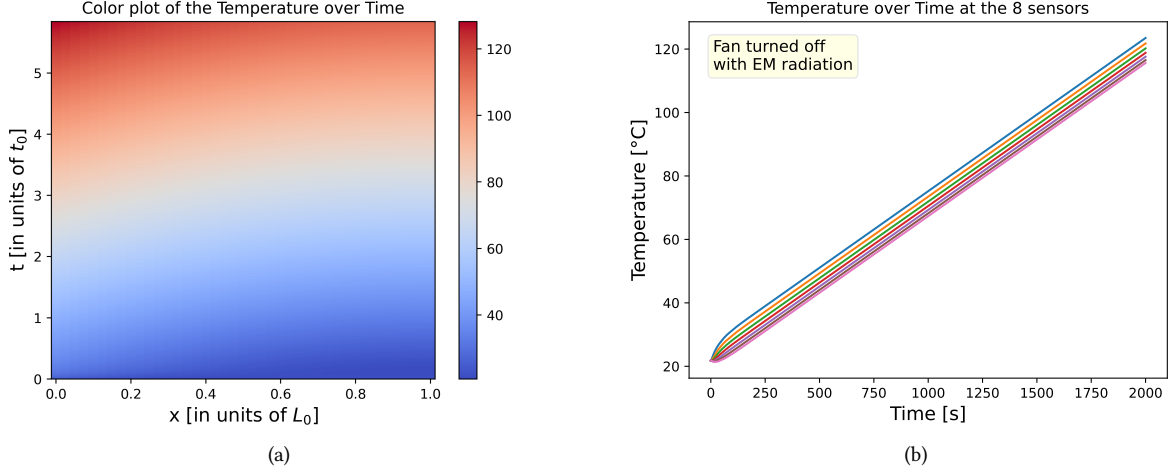


Figure 7: (a) Temperature profile along the Cu bar as a function of time (fan turned off) according to the simulation results using the forward Euler scheme taking into account black body radiation. The temperatures are shown by a color scale ranging from $T = 22$ °c (blue) to $T = 120$ °c (red). (b) The temperature as a function of time at the 8 sensors.

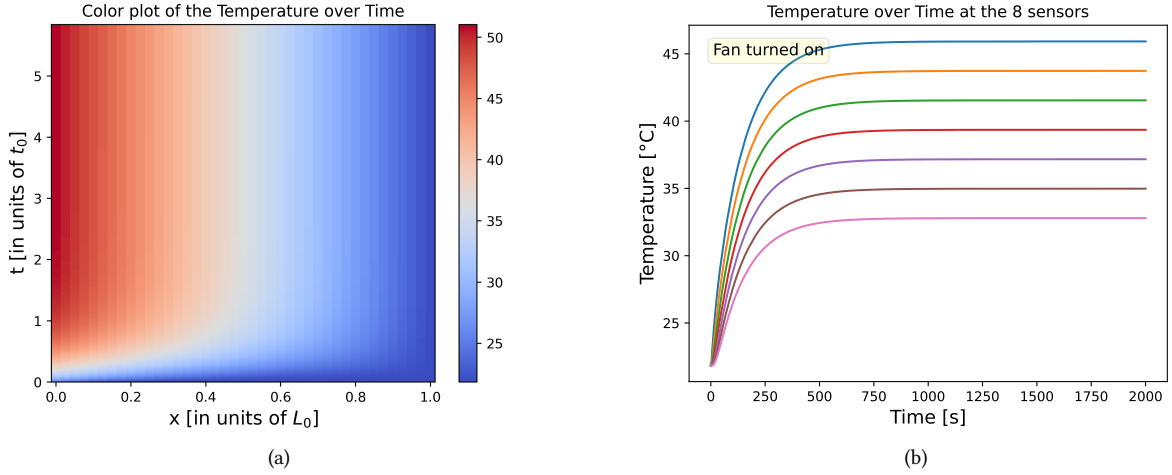


Figure 8: (a) Temperature profile along the Cu bar as a function of time (fan turned on) according to the simulation results using the forward Euler scheme. The temperatures are shown by a color scale ranging from $T = 22$ °c (blue) to $T = 50$ °c (red). (b) The temperature as a function of time at the 8 sensors.

The temperature profile along the Cu bar as a function of time, shown in Figure 8(a), indicates that the temperature is highest at the start of the bar and decreases along the bar when the fan is turned on. Figure 8(b) shows a comparison of the temperature at the 8 sensors as a function of time between the simulation results and experimental data (Figure 5). The simulation results are in close agreement with the experimental data. However, the temperature plateaus after about 500 s in the simulation, whereas the experimental data shows an ongoing increase in temperature. Additionally, the maximum temperature reached in the simulation is 45 °c, whereas the experimental data shows a maximum temperature of 55 °c. These deviations are likely due to experimental uncertainty, errors in simulation parameters, and the neglect of phenomena such as heat convection and black body radiation.

To account for the effect of black body radiation in the code, we can set $f = f_{\text{EM_radiation}}$ instead of 0 in the code (1). The reduced sink term is calculated the same as previously using Equation 9.

Figure 9(a) and 9(b) show that incorporating black body radiation into the simulation decreases the maximum temperature reached from 45 °c to 40 °c. However, the agreement between the simulation results and the experimental data is still not ideal, suggesting that additional factors, such as heat convection, may need to be considered to accurately simulate the temperature. Despite this, the results indicate that the simulation still captures the overall trend of the temperature profile with the fan turned on.

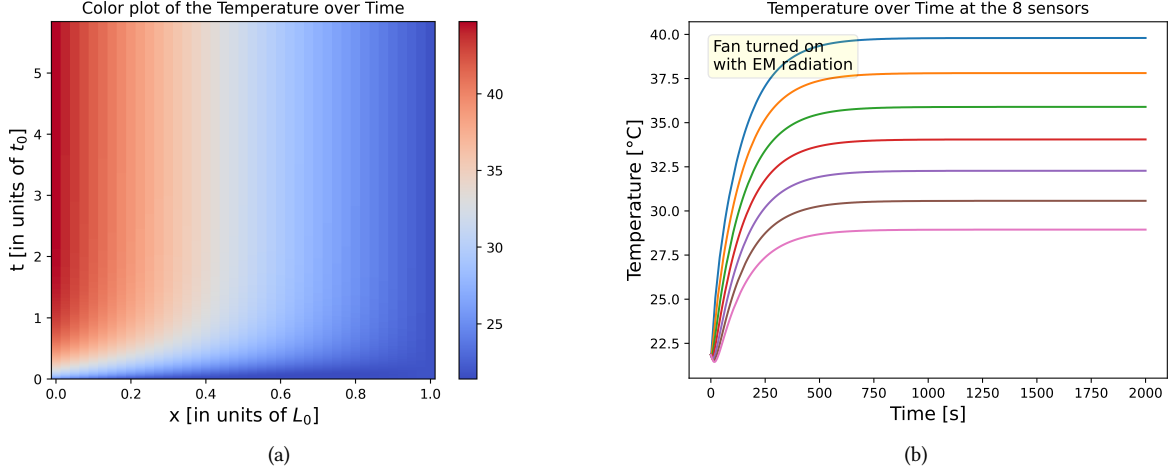


Figure 9: (a) Temperature profile along the Cu bar as a function of time (fan turned on) according to the simulation results using the forward Euler scheme taking into account black body radiation. The temperatures are shown by a color scale ranging from $T = 22$ °c (blue) to $T = 45$ °c (red). (b) The temperature as a function of time at the 8 sensors.

6.2 Backward Euler Method

We then consider the case of the Backward Euler scheme, with three cases: the fan turned off, the fan turned on, and considering a Robin boundary condition. For each case, we present a color plot of the temperature profile along the bar, and a plot of the temperature at the 8 sensors to compare it with the experimental data.

Case 1: Fan turned off

In this case, we applied the Neumann boundary condition at $x = 0$ and $x = L$, with the heat flux at $x = 0$ being $\phi_0 = 60000 \text{ W m}^{-2}$ and at $x = L$ it is set to $\phi_L = 0$. The color plot of the temperature profile along the Cu bar as a function of time is shown in Figure 10(a) and the plot of the temperature at the 8 sensors as a function of time is shown in Figure 10(b).

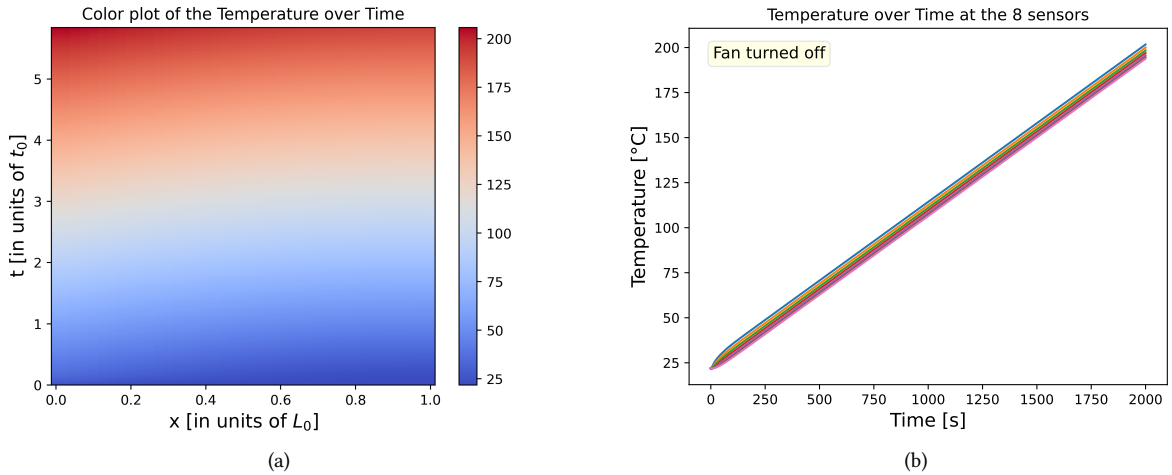


Figure 10: (a) Temperature profile along the Cu bar as a function of time (fan turned off) according to the simulation results using the backward Euler scheme. The temperatures are shown by a color scale ranging from $T = 22$ °c (blue) to $T = 200$ °c (red). (b) The temperature as a function of time at the 8 sensors.

The results of the backward Euler scheme show that the temperature profile along the Cu bar and the temperature at the 8 sensors are similar to the results obtained using the forward Euler method (see section 6.1). This suggests that the backward Euler scheme also does not produce accurate results for the case of the fan turned off, likely due to neglecting heat convection and black body radiation.

However, including the term $u_i^{4^{n+1}}$ in the sink term $\tilde{f}_i^{n+1}$ in Equation 6 to account for the black body radiation becomes very challenging as it is not possible to include the power 4 term in the matrix in the Thomas algorithm. This makes the system in Equation 6 difficult to solve.

Case 2: Fan turned on

As in the previous cases, the Neumann boundary condition is applied at $x = 0$ with the heat flux given by $\phi_0 = 60000 \text{ W m}^{-2}$. At $x = L$, the temperature is kept constant with a Dirichlet boundary condition of $T_L = T_0 = 295 : \text{K}$. The temperature profile along the Cu bar and the temperature at the 8 sensors, as a function of time, are shown in Figure 11(a) and Figure 11(b) respectively.

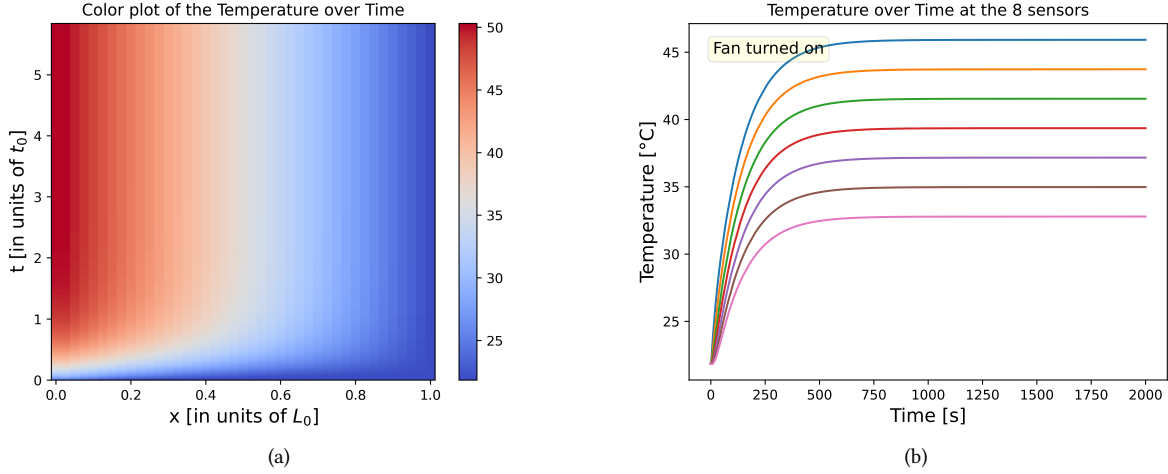


Figure 11: (a) Temperature profile along the Cu bar as a function of time (fan turned off) according to the simulation results using the backward Euler scheme. The temperatures are shown by a color scale ranging from $T = 22^{\circ}\text{C}$ (blue) to $T = 50^{\circ}\text{C}$ (red). (b) The temperature as a function of time at the 8 sensors.

From the figures, it can be seen that the results of the backward Euler scheme are similar to those obtained using the forward Euler method (as seen in section 6.1). This suggests that the backward Euler scheme is able to capture the overall trend of the temperature profile with the fan turned on.

Case 3: Robin boundary condition

In this case, a Neumann boundary condition with a heat flux of $\phi_0 = 60000 \text{ W m}^{-2}$ is applied at $x = 0$. At $x = L$, a Robin boundary condition is used, where the heat flux is defined as $\phi_L = -\lambda \frac{\partial T}{\partial x} = h(T_L - T_{room}) = h(T_L - T_0)$ for all t , with h representing the convection coefficient at the bar-air interface.

It's important to note that the convection coefficient at the copper-air interface is influenced by multiple factors such as air velocity and temperature, copper surface roughness, and system geometry. The coefficient can range from 5 to $200 \text{ W m}^{-2} \text{ K}^{-1}$.

The temperature profile along the Cu bar and the temperature at the 8 sensors as a function of time is shown in Figures 12(a) and 12(b), respectively.

From Figure 12(a), we can observe that the temperature gradually increases across the entire bar with higher temperatures close to the heat source at the start of the bar. Figure 12(b) exhibits a significant improvement in agreement with the experimental data (see Figure 5) when compared to both the forward and backward Euler methods with a Neumann boundary condition at $x = L$. The plot now exhibits curvature instead of a straight line, and the maximum temperature is 100°C , closer to the maximum temperature observed in the experimental data, which was 80°C . These results indicate that considering a convection coefficient of $h = 200 \text{ W m}^{-2} \text{ K}^{-1}$ at the end of the bar significantly improves the accuracy of the simulation results.

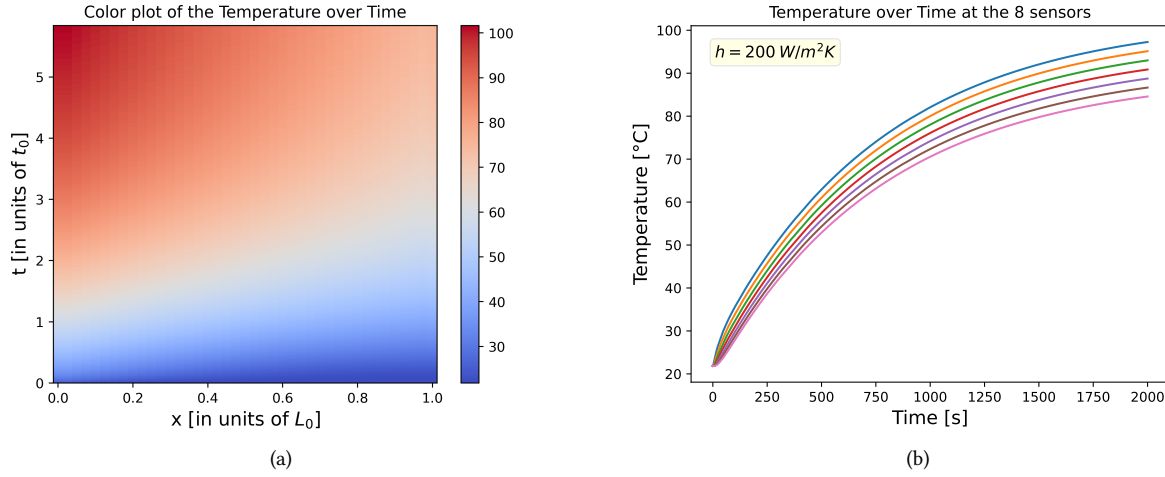


Figure 12: (a) Temperature profile along the Cu bar as a function of time (fan turned off) according to the simulation results using the backward Euler scheme with a Robin boundary condition at $x = L$ taking $h = 200 \text{ W m}^{-2} \text{ K}^{-1}$. The temperatures are shown by a color scale ranging from $T = 22^\circ\text{C}$ (blue) to $T = 100^\circ\text{C}$ (red). (b) The temperature as a function of time at the 8 sensors.

7 Conclusion

The simulation of heat transfer in a copper bar with a Neumann boundary condition at one end and a Dirichlet or Robin boundary condition at the other has been performed using the Forward Euler and Backward Euler methods. The simulation was done for three cases: (1) fan turned off, (2) fan turned on, and (3) fan turned off with a Robin boundary condition at $x = L$. The simulation results were compared with experimental data.

In the first case, the simulation results using both the Forward Euler and Backward Euler methods failed to capture the overall trend of the temperature profile with the fan turned off. The use of black body radiation in the Forward Euler method showed slightly improved agreement with the experimental data.

In the second case, the simulation results using the Backward Euler method showed similar results as the Forward Euler method, with the temperature profile along the copper bar and the temperature at the 8 sensors being captured well by the simulation.

In the third case, when a Robin boundary condition with a convection coefficient increased to 200 was applied at $x = L$, the simulation results showed much improved agreement with the experimental data compared to the first case. This was seen in the curvature of the temperature profile along the copper bar and in the improved match between the simulation and experimental data for the temperature at the 8 sensors.

Overall, these results highlight the importance of considering different boundary conditions and using appropriate methods for accurate simulation of heat transfer in materials, and that the use of a Robin boundary condition at $x = L$ can greatly improve the accuracy of the simulation results.